



A Coupled Statistical and Deterministic Model for Forecasting Climate-Driven Dengue Incidence in Selangor, Malaysia

Xinyi Lu¹ · Su Yean Teh¹  · Hock Lye Koh² · Pei Shan Fam¹ · Chai Jian Tay³

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Abstract

The mosquito-borne dengue virus remains a major public health concern in Malaysia. Despite various control efforts and measures introduced by the Malaysian Government to combat dengue, the increasing trend of dengue cases persists and shows no sign of decreasing. Currently, early detection and vector control are the main methods employed to curb dengue outbreaks. In this study, a coupled model consisting of the statistical ARIMAX model and the deterministic SI-SIR model was developed and validated using the weekly reported dengue data from year 2014 to 2019 for Selangor, Malaysia. Previous studies have shown that climate variables, especially temperature, humidity, and precipitation, were able to influence dengue incidence and transmission dynamics through their effect on the vector. In this coupled model, climate is linked to dengue disease through mosquito biting rate, allowing real-time forecast of dengue cases using climate variables, namely temperature, rainfall and humidity. For the period chosen for model validation, the coupled model can forecast 1–2 weeks in advance with an average error of less than 6%, three weeks in advance with an average error of 7.06% and four weeks in advance with an average error of 8.01%. Further model simulation analysis suggests that the coupled model generally provides better forecast than the stand-alone ARIMAX model, especially at the onset of the outbreak. Moreover, the coupled model is more robust in the sense that it can be further adapted for investigating the effectiveness of various dengue mitigation measures subject to the changing climate.

✉ Su Yean Teh
syteh@usm.my

¹ School of Mathematical Sciences, Universiti Sains Malaysia, 11800 USM, Pulau Pinang, Malaysia

² Jeffrey Sachs Center On Sustainable Development, Sunway University, 47500 Bandar Sunway, Selangor, Malaysia

³ Centre for Mathematical Sciences, Universiti Malaysia Pahang Al-Sultan Abdullah, 26300 Gambang, Pahang, Malaysia

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1 Introduction

Dengue is a mosquito-borne viral infection caused by a virus called dengue virus. It is rampant in urban and semi-urban areas with tropical and sub-tropical climates with variations in risk subject to local climate conditions as well as social and environmental factors. It remains a major public health concern, particularly in Asia where 70% of the global burden of the disease is accounted for (Bhatt et al. 2013). Globally, World Health Organization (WHO) reported an eightfold rise in dengue cases over the last two decades (2000–2019). Malaysia also exhibited similar rising trend in dengue cases with a surge in cases every 4 to 5 years. Although the number of dengue cases seems to decrease in 2020 and 2021, probably due to the pandemic lockdown effect, the number of dengue cases surged again in 2022, following the pre-lockdown trend. The Malaysian Government has introduced various efforts and control measures to combat dengue (Mudin 2015) but the increasing trend of dengue cases persisted. In the absence of antiviral treatment for dengue in Malaysia, early detection and vector control remain the main mechanisms used to curb the outbreak. In Malaysia, vector control mainly involves the use of chemical insecticides to kill the larvae and adult mosquitoes and elimination of mosquito breeding sites. Novel mosquito control approach is also being experimented (Nazni et al. 2019; Ismail et al. 2021).

The ability to detect an outbreak of dengue in advance would assist in timely and cost-effective deployment of vector control measures. In recent years, there were attempts to predict dengue cases for localized areas in Malaysia (Jayaraj et al. 2019; Ismail et al. 2022). In particular, Ismail et al. (2022) tested the forecasting ability of Autoregressive Distributed Lag model, Hierarchical Forecasting, and machine learning methods [Artificial Neural Network, Support Vector Machine and Random Forest (RF)] in predicting dengue outbreaks in two towns of Selangor, namely Selayang and Bandar Baru Bangi. In their study, 81 weeks of epidemiological, entomological, and environmental data were used, out of which the first 70 weeks were used for model evaluation/testing and the rest for measuring the model's performance. They found that the RF produces the best result with 95% accuracy but noted that the method is very costly and time-consuming since entomological data are needed. However, they noted that by excluding the entomological factor, the accuracy only dropped by 3%. Hence, they proposed the integration of the RF algorithm into the e-Dengue system developed by Ministry of Health (MOH) Malaysia to forecast dengue cases up to 4 weeks in advance using the epidemiological data readily available in the e-Dengue system and the environmental data through automated mobile weather stations (~ USD500 each) located at the dengue hot spots. Real-time dengue forecasting has its challenges. Ideally, an abundance of reliable, refined and continuously updated epidemiological, entomological and environmental data are required for accurate prediction. But for Malaysia, obtaining such data can be a challenge for researchers due to data collection cost and data sharing policy.

Climate variables such as temperature, precipitation and humidity can affect dengue transmission dynamics through their effect on vector populations. Temperature can

affect the development rates, mortality and behaviour of mosquitoes (Rueda et al. 1990; Tun-Lin et al. 2000) while variability in precipitation can affect the availability and accessibility of habitats for the offspring of mosquitoes (Morin et al. 2013). Moreover, when temperature interacts with precipitation, the two climate factors can regulate evaporation and thereby influencing the mosquito activity. Considering the potential correlation between climate and dengue cases, the potential use of climate variables, namely temperature, humidity, and precipitation, to forecast dengue incidence in Malaysia was explored in our study. Many studies have attempted to forecast dengue cases using either time series (Cortes et al. 2018; Pham et al. 2020; Costa et al. 2022; Vyhmeister et al. 2022) or mathematical models (Abidemi and Aziz 2020; Fatmawati 2021; Lizarralde-Bejarano et al. 2022; de Araújo et al. 2023). In time series models, a link between climate variables and dengue cases is often established so that future dengue cases can be predicted based on the forecasted climate. But the effect of planned intervention measures cannot be predicted using such time series models. To overcome this, a dengue forecast model that incorporated the elements of a deterministic model and a statistical model is proposed. This coupled model can not only be used to forecast the number of dengue cases but also to investigate the potential change in dengue cases subject to specific vector control measures, especially in the changing climate. In this study, the deterministic SI-SIR (Susceptible-Infective for vector populations; Susceptible-Infective-Recovered for human populations) model is coupled with the statistical Autoregressive Integrated Moving Average Exogenous Variable (ARIMAX) model to enable prediction of dengue cases based on forecasted temperature, humidity, and precipitation. The model was calibrated and validated using the dengue and meteorological data for the state of Selangor in Malaysia. Selangor was chosen as the study site because this state has the highest number of dengue cases and its meteorological data such as average temperature, relative humidity and accumulated precipitation were readily available.

The meteorological and dengue incidence data for the state of Selangor and the details of model are presented next in Materials and Methods. In the Results section, the results of Granger causality test that verified the usefulness of climate variables in predicting mosquito biting rates estimated from dengue incidence are first presented. Next, the forecasting ability of the coupled SI-SIR and ARIMAX model is analysed and discussed. For this purpose, about 5.4 years of meteorological and dengue data were used to train the model before using the trained model to forecast dengue cases in the subsequent weeks. In Discussion, the importance of this study in mitigating dengue outbreak and several limitations of the proposed method are discussed. Lastly, the findings of the study are summarized in the Conclusions section.

2 Materials and Methods

The sequence of the analyses performed in this study is generally illustrated in Fig. 1. The biting rate time series was first estimated by fitting the simulated result of SI-SIR model to the dengue data. Then, along with the time series of average temperature, relative humidity and accumulated precipitation, the four time series were tested for stationarity through Augmented Dickey Fuller (ADF) test. After the stationarity was

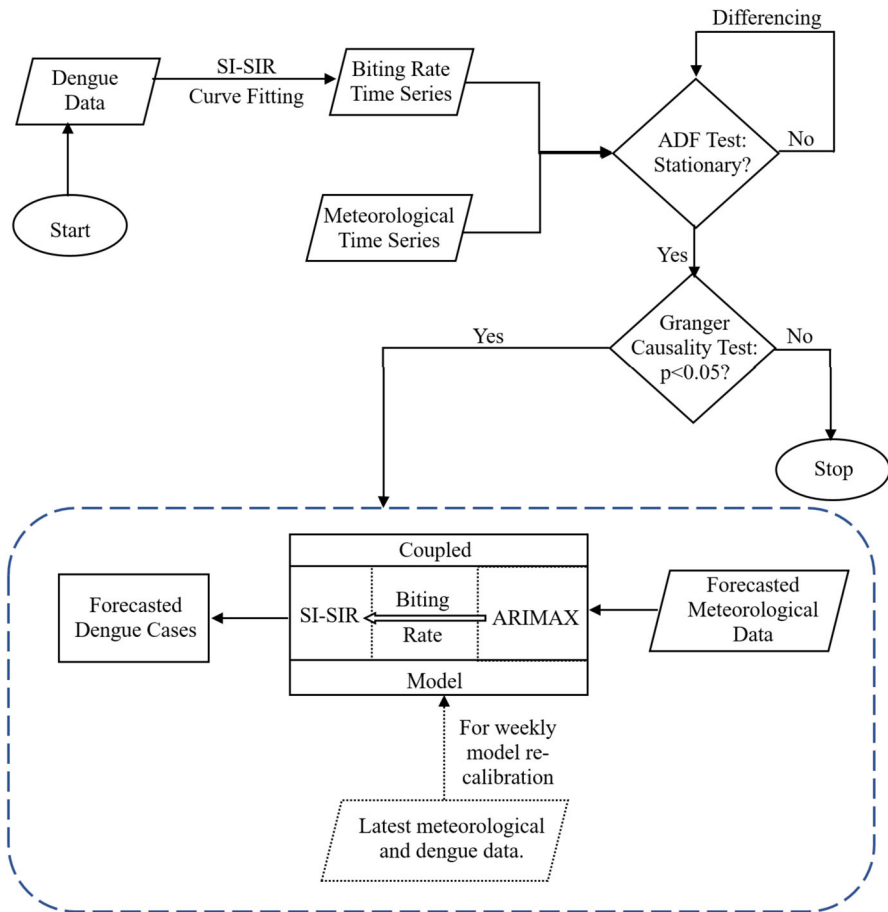


Fig. 1 Flow chart denoting the sequence of analyses performed in this study

confirmed, Granger causality tests were performed (i) between average temperature and average biting rate, (ii) between relative humidity and average biting rate and (iii) between accumulated precipitation and average biting rate. After verifying the one-way Granger causality, by means of Granger causality test, that the three climate factors are helpful for forecasting the average biting rate, the coupled ARIMAX and SI-SIR model can then be used to forecast the number of dengue cases in the following weeks based on the forecasted meteorological data. In the following sections, we first present and discuss the dengue and meteorological data for the state of Selangor in Malaysia. The dengue data of two districts, namely Gombak and Hulu Langat, that were used to further assess the viability of the coupled model are also included. Then the coupled statistical ARIMAX and deterministic SI-SIR model is presented.

2.1 Dengue and Meteorological Data

Selangor is one of the densely populated and highly urbanized states located in the west coast of Peninsular Malaysia. Out of the 13 states and 3 federal territories in Malaysia, the state of Selangor has consistently recorded the highest number of dengue cases, nationally. Hence, Selangor was chosen as the study site in this study.

In this study, the dengue data, specifically the total number of weekly reported dengue cases in the Selangor state from January 2012 to December 2020, were secured in 2021 from the MOH Malaysia through application submitted to the National Medical Research Register. The dengue cases were determined through a standardized diagnostic process, containing the steps of clinical evaluation, blood tests, and laboratory confirmation (MOH-AMM 2015). Local health authorities report the number of confirmed dengue cases to the MOH, who compiled and maintained the data.

As shown in Fig. 2, the reported number of dengue cases is generally higher from 2014 to 2019. Before 2014, the reported number of dengue cases remains at a relative lower level compared the years after 2014. This could be due to the change in the methods used in diagnosing dengue cases (MOH-AMM 2015). The reported number of dengue cases significantly dwindled after 2019, possibly a result of the nationwide lockdown due to the COVID-19 pandemic. On 18 March 2020, the Malaysian Government implemented the Movement Control Order (MCO) to break the COVID-19 transmission chain in Malaysia (Teh et al. 2021; Yong and Sia 2021). During the implementation of MCO in Malaysia, mass movement and gatherings across the country were suspended and many non-essential industries including construction had to shut down. It should be noted that construction sites have been identified as one of the hotspot locations for mosquito breeding due to bad housekeeping habits (MOH 2011;

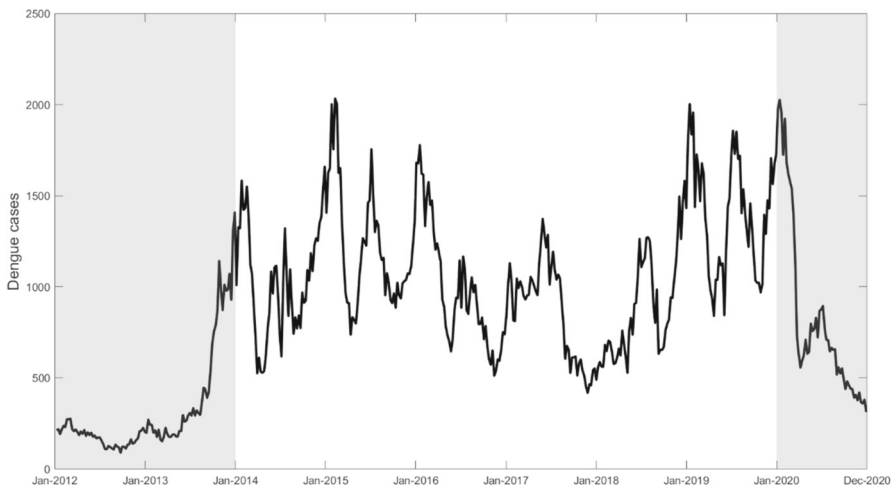


Fig. 2 Reported dengue cases from Jan 2012 to Dec 2020. The data in the shaded periods (from Jan 2012 to Dec 2013 and from Jan 2020 to Dec 2020) are excluded from the analysis to minimize the anomalous factors affecting the reported number of dengue cases

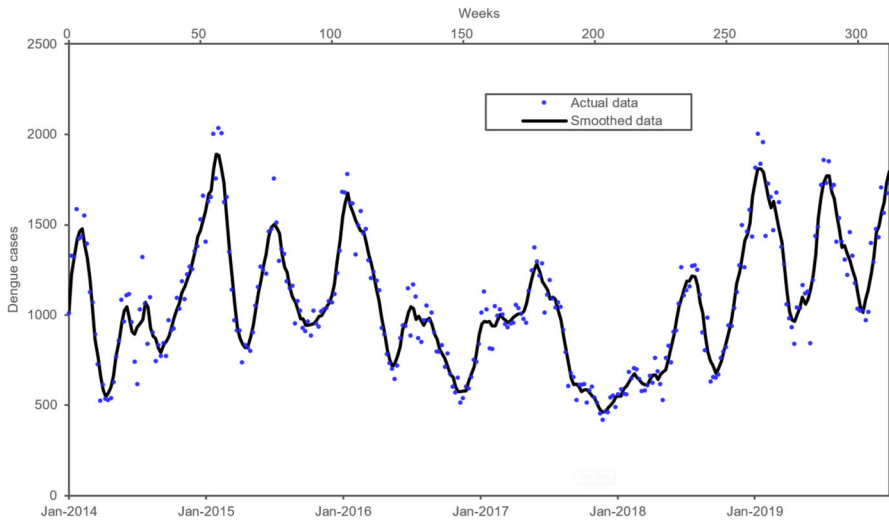


Fig. 3 Actual (dot) and smoothed (line) dengue data for Selangor from Jan 2014 to Dec 2019

Basar et al. 2017). Further, restrictions on movement of people could lead to reduced access to blood meals and lower vector densities, particularly of *Aedes aegypti*, which would contribute to lower dengue incidence (Surendran et al. 2022). Because of these anomalous factors affecting the reported number of dengue cases before 2014 and after 2019, only dengue data from 2014 to 2019 are considered in this study. The weekly dengue data in Fig. 2 are smoothed to determine the underlying trend of data points for statistical analysis. Figure 3 shows both the actual and smoothed dengue data from 2014 to 2019 for Selangor. Here, the moving average filter, which is optimal for reducing random noise while retaining a sharp step response, was used to obtain the smoothed data.

The smoothed and raw dengue data for the two districts Gombak and Hulu Langat, used in the comparison study with the model proposed by Ismail et al. (2022), are shown in Fig. 4.

Temperature, humidity and precipitation are the three main climate variables that have been reported by various researchers to have an effect on dengue transmission (Hii et al. 2009; Chen et al. 2010; Gharbi et al. 2011; Descoux et al. 2012). The climate condition governed by these three variables will affect the population size, maturation period, feeding characteristics and survival rates of mosquitoes (Dom et al. 2013), which will in turn affect the dengue transmission rate. Therefore, the climate variables, namely temperature, humidity and precipitation were considered in this study. Obtained from the Malaysian Meteorological Department (MetMalaysia), the daily averaged temperature, relative humidity and accumulated precipitation used in this study for the state of Selangor were recorded by the Sepang/Kuala Lumpur International Airport monitoring station. To coincide with the weekly dengue data, the meteorological data were processed into weekly intervals. The average temperature

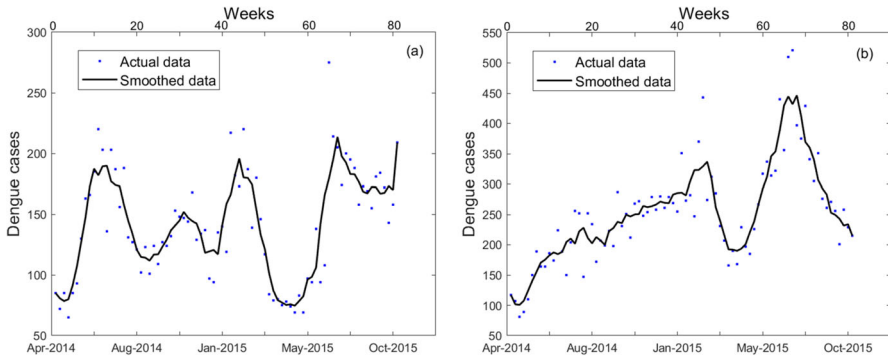


Fig. 4 Actual (dot) and smoothed (line) dengue data for **a** Gombak and **b** Hulu Langat from Apr 2014 to Oct 2015

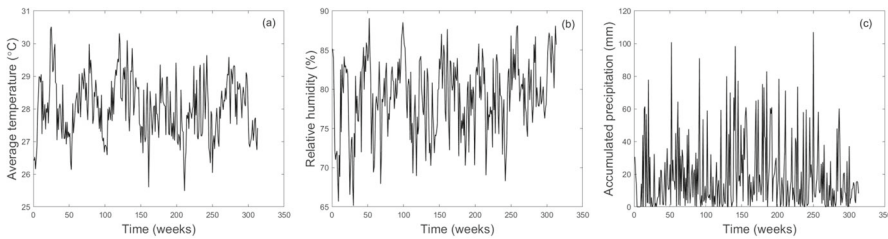


Fig. 5 Average temperature **a**, relative humidity **b** and accumulated precipitation **c** for Selangor from years 2014 to 2019

and relative humidity in a week were obtained by calculating the arithmetic mean values of daily temperature and relative humidity while the precipitation in a week was obtained by calculating the accumulated precipitation for that week. Figure 5 shows the calculated average temperature, relative humidity and accumulated precipitation in weeks from January 2014 to December 2019.

2.2 Stationary Test

Granger causality test can be used to test the correlation between biting rate and climate variables (temperature, rainfall, and humidity). Before Granger causality test and forecasting are performed, the stationarity of the series must be checked first. In most time series models, the parameters related to auto-regressive (AR) and moving average (MA) are considered constants. If the time series is not stationary, the parameters to be estimated are going to be functions of time, which violate the basic assumption of the models. Therefore, the stationarity of the time series must be tested before applying these time series analysis techniques. It is observed from Fig. 5 that the climate variables largely fluctuate around a mean value, which implies that the sequences might be stationary. The Augmented Dickey Fuller test (ADF Test) is a widely used statistical

test that belongs to a category of tests called unit root test, which is a proper method to test whether a given time series is stationary or not.

2.3 Granger Causality Test

Malaysia is a tropical country with slightly varied temperature, high humidity and abundant precipitation, as evident from Fig. 5. To know whether these climate variables are useful in predicting the mosquito biting rate, Granger causality test (Granger 1969) was used. It should be noted that the word ‘causality’ is not really the same as the concept of cause and effect that is commonly understood. Here, the causality means ‘Granger-cause’, which means one variable precedes another in the time series. Hence, it is suitable for testing whether one time series is helpful for forecasting another. Granger causality test is often applied when two conditions are satisfied. The first condition is that the forecasting performance is the focus, not the theoretical model behind the performance. As each of the implicit and explicit restrictive assumptions for identifying the Granger causal relationship is unlikely to hold in practice (Shojaie and Fox 2022), Granger causality focuses more on whether a time series can be predicted using all available information than if the information apart from another time series had been used. This is what the first condition emphasizes on. The second condition is that the stationary data for the test are available. For this study, Eq. (1) can be used.

$$Y(t) = \alpha_0 + \sum_{i=1}^p \alpha_i X(t-i) + \sum_{i=1}^q \beta_i Y(t-i) + \mu(t) \quad (1)$$

Here, α_0 is a constant, α_i denotes the coefficient parameters of X , β_i denotes the coefficient parameters of Y , $\mu(t)$ denotes the white noise, while p and q denote the maximum time lag of X and Y respectively. Notably, the Vector Autoregressive (VAR) model is often employed as a tool to access Granger causality. The estimations of VAR models with lagged values are then used to perform Granger causality tests. A comparative analysis was conducted utilizing various information criteria, including the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). AIC is based on a forecasting perspective while BIC is based on the fitting perspective (Chakrabarti and Ghosh 2011). In this study, the AIC is used as the basis for lag order selection (Ozcicek and McMillin 1999). The lag order that minimizes the selected information criterion was chosen. To test whether the climate variables are useful for predicting the average biting rate, the Granger causality test was performed between (i) average temperature and average biting rate, (ii) relative humidity and average biting rate, and (iii) accumulated precipitation and average biting rate.

2.4 Coupled ARIMAX and SI-SIR Model

The deterministic SI-SIR model and statistical ARIMAX model were coupled in MATLAB (2022) and used to forecast the dengue incidence in Selangor. The ARIMAX module in the coupled model first predicts the average biting rate per week for the next

few weeks (i.e., the forecast period) based on the known or projected weekly-averaged climate variables within the forecast period. Then the predicted average biting rate is fed into the SI-SIR module in the coupled model to forecast the number of dengue cases within the forecast period. To improve the forecast, the model is continuously re-calibrated every week to revise the predicted average biting rate by incorporating the latest meteorological and dengue data. To assess the performance of the coupled model, the magnitude of relative error (MRE) between the forecasted number of dengue cases and the reported number of dengue cases is calculated.

2.4.1 SI-SIR Model

In most dengue research, the vector-host models that track the dynamics of both mosquitoes and humans were used to study the transmission of dengue disease. Among those vector-host models, the SI-SIR model, which was comprehensively discussed by Derouich et al. (2003), was commonly used. The SI-SIR model assumes homogeneous mixing of human and mosquito populations. In such compartmental models, each population is divided into classes, and individuals of a population pass from one class to another at an appropriate rate. A SI-SIR model was previously validated by Tay et al. (2022) to be suitable for illustrating the dengue transmission dynamics in Kuala Lumpur and Selangor, Malaysia. This model indicated the critical role of the dengue infection rate during the dengue transmission process and captured the pattern of dengue cases in both study sites.

The system of ordinary differential equations (ODEs) for the SI-SIR model used in this study is given in Eq. (2). The mosquito population is divided into susceptible mosquitoes (V_S) and infected mosquitoes (V_I) while the human population is divided into three compartments, which are susceptible human (H_S), infected human (H_I) and recovered human (H_R). The total number of humans is $N_H = H_S + H_I + H_R$ while the total number of mosquitoes is $N_V = V_S + V_I$. The definition and unit of the parameters in SI-SIR model are listed in Table 1. Values of the parameters used in the simulation study are also provided together with the source, from which the values were obtained.

$$\begin{aligned}
 \frac{dV_S}{dt} &= \mu_V N_V - B\beta_{HV} \frac{H_I}{N_H} V_S - \mu_V V_S \\
 \frac{dV_I}{dt} &= B\beta_{HV} \frac{H_I}{N_H} V_S - \mu_V V_I \\
 \frac{dH_S}{dt} &= \mu_H N_H - B\beta_{VH} \frac{V_I}{N_H} H_S - \mu_H H_S \\
 \frac{dH_I}{dt} &= B\beta_{VH} \frac{V_I}{N_H} H_S - \gamma_H H_I - \mu_H H_I \\
 \frac{dH_R}{dt} &= \gamma_H H_I - \mu_H H_R
 \end{aligned} \tag{2}$$

The average biting rate for Selangor is estimated by fitting the SI-SIR model (2) to the dengue data from 2014 to 2019. The fourth equation in (2) is deemed to be

Table 1 Definition, unit and value of the parameters in the SI-SIR model used in this study

Parameter	Definition	Value	Unit	Source
V_S	Susceptible mosquito population	–	Capita	–
V_I	Infected mosquito population	–	Capita	–
H_S	Susceptible human population	–	Capita	–
H_I	Infected human population	–	Capita	–
H_R	Recovered human population	–	Capita	–
N_V	Mosquito population	$2 \cdot N_H$	Capita	–
N_H	Human population	$\frac{1}{2} \cdot N_V$	Capita	–
μ_V	Natural mortality rate of mosquito	0.7	Week ⁻¹	Andraud et al. (2012)
μ_H	Natural birth/mortality rate of human	$\frac{1}{75 \times 52}$	Week ⁻¹	Tay et al. (2022)
B	Average biting rate	Curving fitting	Week ⁻¹	–
β_{HV}	Probability of human-to-mosquito transmission	0.5	–	de Lima et al. (2016)
β_{VH}	Probability of mosquito-to-human transmission	0.75	–	Derouich et al. (2003)
γ_H	The recovery rate of infected human	0.7	Week ⁻¹	Andraud et al. (2012)

the most suitable for deriving the average biting rate because the data on infected cases are available. The Euler's Method (also called Forward Euler method), which is the most basic explicit method for solving ordinary differential equations with given initial values, is used to discretize the equation. When using the Euler's Method, a differential equation $d\omega/dt = f(t, \omega)$ with an initial condition $\omega(t_0) = \omega_0$ is considered. Then a successive approximation of the equation can be given by $\omega_{i+1} = \omega_i + h \cdot f(t_i, \omega_i)$, where h indicates the step size. Following this, the fourth equation in Eq. (2) is discretised into $H_{I,i+1} = H_{I,i} + \Delta t [B_i \beta_{VH} \frac{V_{I,i}}{N_H} H_{S,i} - \gamma_H H_{I,i} - \mu_H H_{I,i}]$, which can be re-arranged to form Eq. (3).

$$B_i = \left(\frac{H_{I,i+1} - H_{I,i}}{\Delta t} + \gamma_H H_{I,i} + \mu_H H_{I,i} \right) \frac{N_H}{\beta_{VH} V_{I,i} H_{S,i}} \quad (3)$$

The ARIMAX model uses (3) to fit the dengue data, enabling the calculation of the average biting rate to obtain a sequence of the rate as a dependent variable in the ARIMAX model. The time series of the three climate variables serve as exogenous variables for the ARIMAX model. By estimating the coefficients in the ARIMAX

model using this combined dataset, future average biting rates can be forecasted using the estimated ARIMAX formula, the process of which is detailed in Sect. 2.4.2.

In our study, the human population N_H used is 6,313,767, an average of Selangor's population from 2014 to 2019. The infected human population (H_I) represents the weekly total dengue cases. Initial value of H_I is set as the total dengue cases of first week and the initial value of recovered human population (H_R) is set as 0. Hence, the initial value of susceptible human population (H_S) can be calculated as $N_H - H_I - H_R$. Similarly, after the initial value of infected mosquito population (V_I) being set, the initial value of susceptible mosquito population (V_S) is calculated as $N_V - V_I$. Subsequently, the estimated biting rate is used to calculate V_S , V_I , H_S , H_I , and H_R in Eq. (4).

$$\begin{aligned}
 V_{S,i+1} &= V_{S,i} + \Delta t \left[\mu_V N_V - B_i \beta_{HV} \frac{H_{I,i}}{N_H} V_{S,i} - \mu_V V_{S,i} \right] \\
 V_{I,i+1} &= V_{I,i} + \Delta t \left[B_i \beta_{HV} \frac{H_{I,i}}{N_H} V_{S,i} - \mu_V V_{I,i} \right] \\
 H_{S,i+1} &= H_{S,i} + \Delta t \left[\mu_H N_H - B_i \beta_{VH} \frac{V_{I,i}}{N_H} H_{S,i} - \mu_H H_{S,i} \right] \\
 H_{I,i+1} &= H_{I,i} + \Delta t \left[B_i \beta_{VH} \frac{V_{I,i}}{N_H} H_{S,i} - \gamma_H H_{I,i} - \mu_H H_{I,i} \right] \\
 H_{R,i+1} &= H_{R,i} + \Delta t [\gamma_H H_{I,i} - \mu_H H_{R,i}]
 \end{aligned} \tag{4}$$

2.4.2 ARIMAX

Statistical models, especially time series models, are widely used in the study of epidemic diseases. Among the time series models, the Autoregressive Integrated Moving Average (ARIMA) model is frequently used for forecasting purpose. The ARIMAX model was used to predict the average biting rate in this study, in which climate variables are included as exogenous variables.

ARIMA is one of the models that interprets a given time series based on its own past values to forecast future values. When exogenous variables are also considered, they constitute the ARIMAX model, which can be expressed as Eq. (5).

$$Y_t = \delta + \sum_{k=1}^K \eta_k X_k + \sum_{p=1}^P \phi_p Y_{t-p} + \sum_{q=0}^Q \theta_q \varepsilon_{t-q} \tag{5}$$

Here, δ is a constant; η_k denotes the coefficient parameters of the exogenous sequence X_k ; ϕ_p denotes the coefficient parameters of the dependence sequence Y , and θ_q denotes the coefficient parameters of residuals ε ; K represents the total count of exogenous variables and k the k th exogenous variable, while P and Q denote the maximum time lag of the dependence sequence and residuals respectively. The 'auto.arima' function via R package (v8.21.1; Hyndman et al. 2023) in R Statistical Software (v4.2.2; R Core Team 2022) was employed to determine the maximum lags in the

model. This function searches through a range of possible lag orders and identifies the one that minimizes a selected information criterion (usually AIC or BIC), without requiring manual intervention (Hyndman and Khandakar 2008). It ensures that the selected time series model is with fewer parameters and is effective. The output generated by the automated function provides a robust and reliable method for ARIMAX model selection and achieves a balance between capturing important temporal dynamics and avoiding the inclusion of noise, which mitigates overfitting concerns. In this study, the time series of climate variables, namely average temperature, relative humidity and accumulated precipitation were regarded as the exogenous sequences in the ARIMAX model while the time series of average biting rate was regarded as the dependence sequence. When estimating the parameters in the model, the current and historical values of the three climate variables along with the average biting rate are used as input to fit Eq. (3). Then the coefficients are obtained by means of maximum likelihood estimation (MLE). With these coefficients, the values of climate variables at time t (any point-in-time in future) are substituted into the estimated ARIMAX formula to forecast the value of average biting rate at time t .

3 Results

Section 3.1 first presents the results of the analysis performed to determine whether climate variables are useful to predict the average biting rate of mosquito. Then the performance of the coupled SI-SIR and ARIMAX model is investigated and discussed in Sect. 3.2 by comparing the forecasted dengue cases with the reported cases. Lastly, the results of the comparison between the coupled model and an existing model are presented in Sect. 3.3.

3.1 Affinity Between Dengue and Climate Variables

In our model, climate is associated with dengue incidence through mosquito biting rate. To obtain the association between the climate variables, namely temperature, humidity and precipitation, and mosquito biting rate, the time series of mosquito biting rate was first obtained using SI-SIR model by curve fitting. Then stationary test was performed to check the stationarity of the time series. Lastly, the correlation between the biting rate and climate variables was tested using the Granger causality test.

3.1.1 Average Biting Rate

The weekly averaged biting rate for the period of 312 weeks (Jan 2014–Dec 2019) was estimated by fitting the SI-SIR model result to both unsmoothed and smoothed dengue data, the result of which is illustrated in Fig. 6. The forecasting performance of the coupled model has been evaluated using the biting rate estimated using unsmoothed data and smoothed dengue data. It was found that the forecast is significantly better using the biting rate from smoothed data. For example, using the unsmoothed or raw data, the average mean magnitude of relative error (MMRE) over 10 rounds for

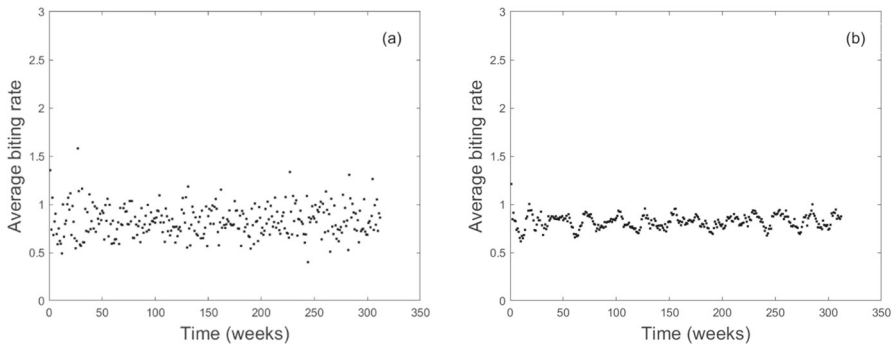


Fig. 6 Estimated average biting rate by curve fitting using **a** unsmoothed dengue data and **b** smoothed dengue data

the forecast period (Weeks 283 to 295) was 17.56%. In contrast, the utilization of smoothed data yielded a substantially lower average MMRE of 8.01% for the same duration. Additionally, the MMRE for each forecast round varied widely when using raw data, ranging from 4.82 to 35.47%, in contrast to the narrower range of 2.28% to 23.09% observed with smoothed data. This finding is in line with the general rule of data analysis that smoothed data could better identify changes in the trend compared to unsmoothed data, which could have more noises that create false signals, as observed in other studies (Goryawala et al. 2020; Shan et al. 2023). Hence, the biting rate calculated by fitting the SI-SIR model result to the smoothed dengue data was used for the subsequent analysis. The estimated average biting rates (Fig. 6b) mostly vary between 0.6 and 1.5 per week. For perspectives, some mosquito biting rates reported in the literature are used as comparison. In Scott et al. (2000), *Aedes aegypti* mosquitoes were collected weekly from houses in rural Thailand for 2 years to study the mosquito feeding habits. Based on their plot of average daily temperature and the average number of blood meals imbibed each week, the biting rates were observed to vary from 1.00 to 1.9 per week for temperature ranging from 25 to 32 °C. Further, Liu-Helmersson et al. (2014) has formulated the equation, $\alpha(T) = 0.0043 T + 0.0943$ per day ($21\text{ }^{\circ}\text{C} \leq T \leq 32\text{ }^{\circ}\text{C}$), for the biting rate α (per day) as a function of temperature T based on data from Scott et al. (2000). This formula gives a biting rate range of about 1.41–1.62 per week for temperature ranging from 25 to 32 °C. More recently, Christofferson et al. (2022) reported a range of daily bites of *Aedes aegypti* for three temperatures based on lab experiments. A range of 0–0.185 per day (0–1.295 per week) was reported for the temperature of 24 °C, 0.038–0.346 per day (0.266–2.422 per week) for 28 °C and 0–0.154 per day (0–1.078 per week) for 32 °C. Hence, the estimated biting rates are comparable to the range reported in the literature.

3.1.2 Stationary Test

ADF test was applied to check the stationarity of average biting rate, average temperature, relative humidity and accumulated precipitation time series. Table 2 lists the value of test statistics and corresponding p values of the ADF test. The result suggests

Table 2 ADF test for average biting rate and climate factors

	Average biting rate	Average temperature	Relative humidity	Accumulated precipitation
Test statistic	− 6.5932	− 5.1580	− 5.0597	− 17.7749
<i>p</i> value	0.0000	0.0000	0.0000	0.0000
Conclusion	stationary	stationary	stationary	stationary

Table 3 Granger causality test between average biting rate and climate factors

Relationship	Average temperature $\Rightarrow$ Average biting rate	Relative humidity $\Rightarrow$ Average biting rate	Accumulated precipitation $\Rightarrow$ Average biting rate
<i>p</i> value	0.0126	0.0000	0.0405
Reverse relationship	Average biting rate $\Rightarrow$ Average temperature	Average biting rate $\Rightarrow$ Relative humidity	Average biting rate $\Rightarrow$ Accumulated precipitation
<i>p</i> value	0.2595	0.1078	0.6996

that all *p* values are less than the significance level 0.05, implying that these four time series are stationary. Therefore, we can proceed with the Granger causality test.

3.1.3 Granger Causality Test

The Granger causality test was used to test whether the climate variables are useful for predicting the average biting rate. In Table 3, the Granger causality between (i) average temperature and average biting rate, (ii) relative humidity and average biting rate, and (iii) accumulated precipitation and average biting rate is denoted by ' $\Rightarrow$ '. For instance, 'average temperature $\Rightarrow$ average biting' means the time series of average temperature Granger causes the time series of average biting rate. The *p* value for all three relationships is less than the significance level 0.05. The reverse causality test between average biting rate and climate factors was also performed. Also shown in Table 3, the *p* values for all the reverse relationships are greater than the significance level 0.05. The results imply that there exists a one-way Granger causality between average biting rate and climate factors. Hence, it can be deduced that the three climate variables, namely average temperature, relative humidity and accumulated precipitation, are useful in predicting the average biting rate for input into the SI-SIR model for forecasting dengue cases.

3.2 Performance of the Coupled SI-SIR and ARIMAX Model

Here, the results for a forecast period of four weeks are presented. But it should be noted that shorter and longer forecast periods have been considered, the result of which will

also be discussed. The coupled SI-SIR and ARIMAX model, trained and calibrated using meteorology and dengue data from Week 0 to Week 282 (approximately Jan 2014–May 2019, Fig. 3), was first used to predict the average biting rate for the next four weeks (i.e., forecast period of 4 weeks). Then the predicted average biting rate for the forecast period was fed into the SI-SIR module in the coupled model to obtain the forecasted number of dengue cases for the next four weeks (Weeks 283–286). For more accurate real-time forecast of dengue cases, the model has to be continuously re-calibrated every week to revise the predicted average biting rate by incorporating the updated meteorological and dengue data. To illustrate this, we performed 10 rounds of forecast with each round incorporating the updated meteorological and dengue data to predict the average biting rate. Table 4 compares the reported and forecasted number of dengue cases and lists the MRE for each week and the MMRE for ten forecast rounds. In forecast round 2, the model included the reported dengue data and recorded climate data for Week 283 in the model training to revise the predicted average biting rate. The revised biting rates were then used to obtain the forecasted dengue cases in Weeks 284–287. Same goes for the subsequent forecast rounds. For example, forecast round 3 was performed using re-calibrated average biting rate up to Week 284 while forecast round 4 was performed using re-calibrated average biting rate up to Week 285. The forecasted number of dengue cases starting Week 283 for up to four weeks for ten rounds of forecast, as listed in Table 4, is also shown graphically in Fig. 7. The error in the first round (forecast round 1) is the highest with MRE ranging from 10.63 to 27.74% with a mean of 23.09%. Starting from Week 283, the number of dengue cases steadily climbs to reach the peak around Week 290 before decreasing. The shift in the trend from decreasing dengue cases before Week 283 to the increasing trend after Week 283 can only be best captured if the model adapts to the shift by continuously updating the forecasted value using updated dengue and climate data.

It can be observed from Table 4 and Fig. 7 that the forecasted number of dengue cases has higher MMREs (11.02–23.09%) in the first few rounds when the outbreak is at the beginning stages. Similar to many forecasting models, it is challenging to predict the early stages of an outbreak. However, as the outbreak progresses, the forecast results in lower MMREs (< 8%). Typically, point forecasting methods are assessed by means of an error measure (Gneiting 2011). MRE is a statistical technique used to measure the error between actual and estimated values relative to the actual values. MMRE is obtained from calculating the average of MRE. In practice, MMRE is widely applied in various fields, including agriculture and engineering, where lower MMRE usually implies a better performance (Fagundes et al. 2013; Omomule et al. 2020; Singal et al. 2020; Kumar et al. 2021; Mahdaviara et al. 2021; Rankovic et al. 2021).

As illustrated in Fig. 7, for each round of forecast, the number of dengue cases was forecasted for four weeks to keep the MMRE low and to allow a convenient illustration of the result. Also, a four-week forecasting period aligns conveniently with approximately 1 month, rendering it more practical for real-time predictions on a monthly basis. Table 5 lists the comparison of MMRE for several forecast periods, namely from 1 to 6 weeks, for 10 rounds of forecast. It can be observed that the average value of MMRE is below 8% for the shorter forecast periods of 1–3 weeks. The average MMRE is slightly above 8% for the forecast period of 4 and 5 weeks and above 10% for forecast periods of 6 weeks. This means that for the period chosen

Table 4 Forecasted number of dengue cases for 10 rounds with magnitude of relative error (MRE) in brackets

Week	Actual dengue cases	Forecast rounds									
		1	2	3	4	5	6	7	8	9	10
283	844	1076 (27.55%)									
284	1193	1066 (10.63%)	1183 (0.87%)								
285	1437	1057 (26.46%)	1232 (14.23%)	1296 (9.83%)							
286	1487	1074 (27.74%)	1318 (11.36%)	1418 (4.61%)	1494 (0.45%)						
287	1719	1356 (21.13%)	1497 (12.92%)	1617 (5.92%)	1704 (0.88%)						
288	1857		1546 (16.74%)	1717 (7.55%)	1858 (0.07%)	1750 (5.76%)					
289	1729			1787 (3.38%)	1992 (15.21%)	1819 (5.18%)	1780 (2.95%)				
290	1851				2098 (13.34%)	1852 (0.06%)	1793 (3.15%)	1778 (3.96%)			
291	1701					1799 (5.78%)	1720 (1.13%)	1698 (0.17%)	1688		
292	1720						1681 (2.28%)	1652 (3.97%)	1637 (4.85%)	1625 (5.52%)	
293	1403							1603 (14.22%)	1583 (12.82%)	1566 (11.62%)	

Table 4 (continued)

Week	Actual dengue cases	Forecast rounds									
		1	2	3	4	5	6	7	8	9	10
294	1536									1536 (0.00%)	1514 (1.42%)
295	1405										1461 (4.00%)
MMRE		23.09%	11.90%	11.02%	4.33%	7.38%	4.20%	2.38%	5.58%	4.61%	5.64%

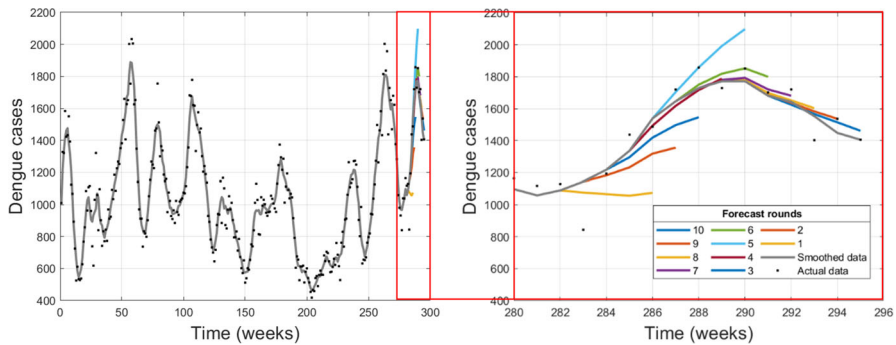


Fig. 7 The forecasted number of dengue cases for up to four weeks for ten rounds of forecast (forecast starts from Week 283) using the coupled ARIMAX and SI-SIR model

Table 5 Comparison of MMRE for various forecast periods

MMRE	Forecast period (week)					
	1	2	3	4	5	6
<i>Forecast round</i>						
1	27.55%	19.09%	21.54%	23.09%	26.20%	29.06%
2	0.87%	7.55%	8.82%	11.90%	14.78%	15.66%
3	9.83%	7.22%	9.12%	11.02%	10.65%	11.12%
4	0.45%	3.19%	4.64%	4.33%	3.65%	4.24%
5	0.88%	0.48%	5.39%	7.38%	10.63%	12.50%
6	5.76%	5.47%	3.67%	4.20%	4.02%	7.32%
7	2.95%	3.05%	2.41%	2.38%	5.24%	5.01%
8	3.96%	2.06%	2.70%	5.58%	4.77%	5.23%
9	0.77%	2.81%	6.14%	4.61%	4.86%	6.31%
10	5.52%	8.57%	6.19	5.64%	6.82%	8.85%
Average	5.85%	5.95%	7.06%	8.01%	9.16%	10.53%

for the validation (Week 283 to 295), the coupled model can forecast 1–2 weeks in advance with an average error of less than 6%, three weeks in advance with an average error of 7.06% and four weeks in advance with an average error of 8.01%.

The forecasting performance of the coupled model was further assessed by comparing its forecasted results to those directly forecasted from the time series using the stand-alone ARIMAX model. The MMREs of the number of dengue cases forecasted by the coupled model and the ARIMAX model for 10 forecast rounds are listed in Table 6. Although in some forecast rounds, the MMRE of the coupled model is larger than the one of ARIMAX, the difference is very small ($\sim 3\%$). On the other hand, in rounds where the MMRE of ARIMAX is larger than MMRE of the coupled model, the difference can be as high as 27.93% to as low as 0.45%. It can be deduced here

Table 6 MMRE of forecasted values by coupled model (ARIMAX + SI-SIR) and ARIMAX denoting a lower average MMRE for the coupled model compared to ARIMAX based on ten rounds of forecast

Model type	MMRE (%)										Average MMRE for 10 rounds
	Forecast round										
	1	2	3	4	5	6	7	8	9	10	
ARIMAX + SI-SIR	23.09	11.90	11.02	4.33	7.38	4.20	2.38	5.58	4.61	5.64	8.01
ARIMAX	20.08	39.83	27.09	14.91	12.34	8.02	2.91	6.40	6.88	6.09	14.46
IDifferenceI	3.01	27.93	16.07	10.58	4.96	3.82	0.53	0.82	2.27	0.45	

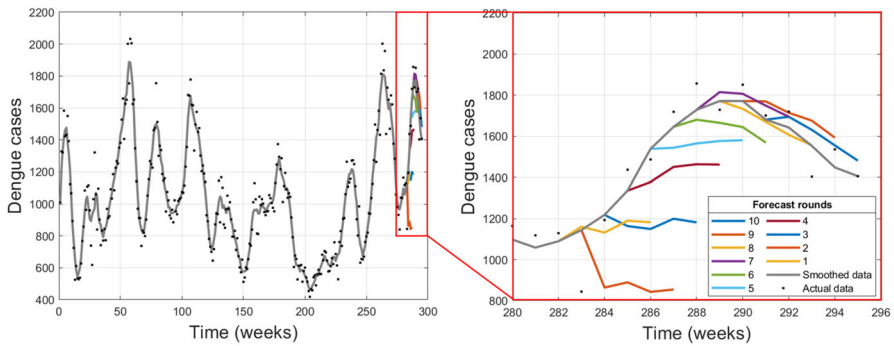


Fig. 8 The forecasted number of dengue cases for up to four weeks for ten rounds of forecast (forecast starts from Week 283) using ARIMAX

that both models were able to forecast at reasonable accuracy when the trend of the outbreak is prominent. But at earlier rounds (onset of the outbreak), the coupled model appeared to perform much better than ARIMAX. Figure 8 shows the forecasted number of dengue cases using ARIMAX for up to four weeks for ten rounds starting from Week 283. Compared to the results of the coupled model illustrated in Fig. 7, it is evident here that the coupled model offers a good if not better forecast, especially at the onset of the outbreak.

To further test the forecasting performance of the coupled model, we randomly choose another validation period (Weeks 204 to 216), the result of which is presented in Fig. 9. In this case, the MMREs mainly stay below 10% except in the ninth (11.04%) rounds. This indicates that the model forecasting performance is reasonably good even for a different validation period.

Furthermore, the values of averaged MMRE using simple ARIMAX (Fig. 10) and the coupled ARIMAX and SI-SIR model (Fig. 11) for three outbreak periods (from March 2018 to August 2019) are compared in Table 7 to assess the forecasting performance for each epidemic curve. The averaged MMRE values for the three outbreak

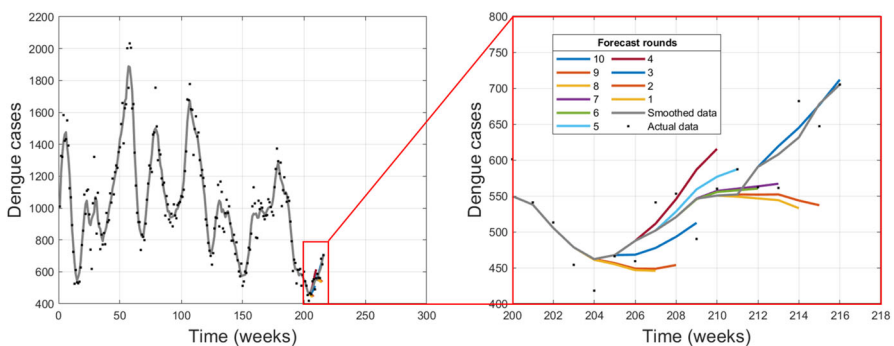


Fig. 9 The forecasted number of dengue cases for up to four weeks for ten rounds of forecast (forecast starts from Week 204) using the coupled ARIMAX and SI-SIR model

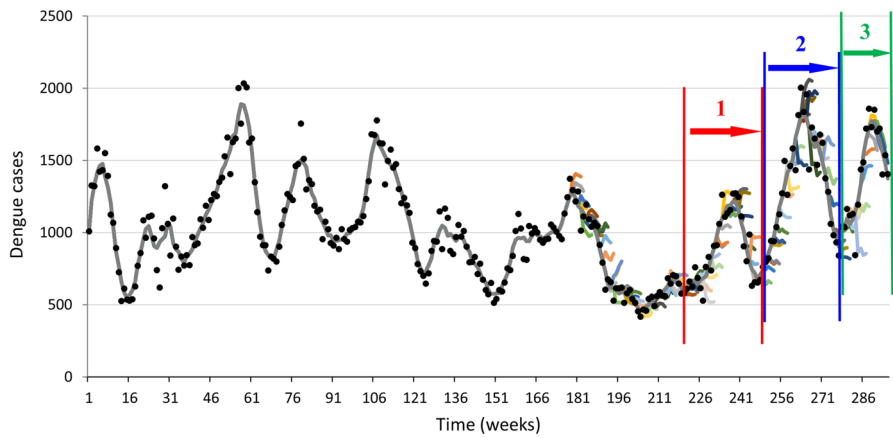


Fig. 10 Real-time prediction of dengue cases for three outbreak periods using ARIMAX

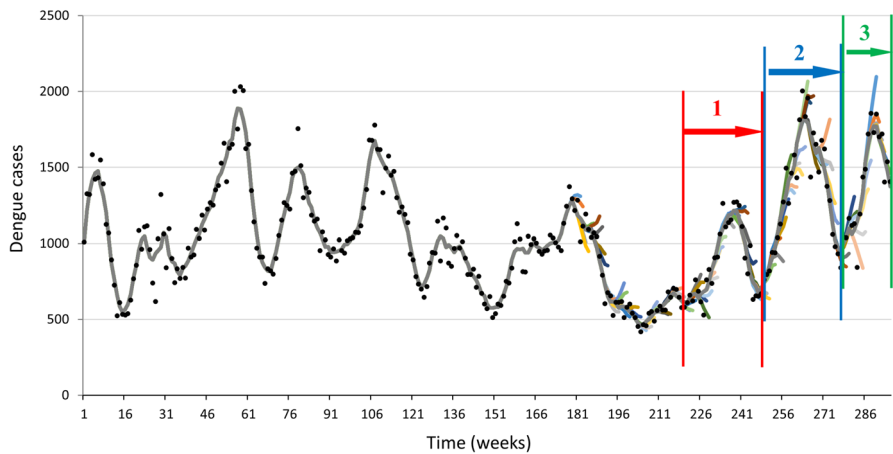


Fig. 11 Real-time prediction of dengue cases for three outbreak periods using the coupled ARIMAX and SI-SIR model

Table 7 Comparison of the values of averaged MMRE for the three outbreak periods indicated in Figs. 10 and 11

Period	Week	Time	Averaged MMRE (ARIMAX)	Averaged MMRE (coupled model)
1	219–248	Mar 2018–Sep 2018	18.21%	11.00%
2	249–277	Oct 2018–Apr 2019	17.87%	9.74%
3	278–295	Apr 2019–Aug 2019	14.44%	9.63%

periods forecasted using the coupled model revolved around 9.63–11% while the averaged MMRE values forecasted using the simple ARIMAX model ranged from 14.44 to 18.21%. This indicates that the prediction capability of the coupled model is fairly consistent in its performance against ARIMAX.

3.3 Comparison with Existing Model

Here, the performance of the proposed coupled model is compared to the performance of the time series models reported in Ismail et al. (2022). With no access to the data for the two study sites (Selayang and Bandar Baru Bangi) considered in their study, the comparison of the model performance was performed using dengue data for the two districts Gombak and Hulu Langat, in which the two study sites are respectively located, and the climate data for Selangor. The coupled model forecasted number of dengue cases in Gombak and Hulu Langat for up to four weeks for fourteen rounds of forecast are shown in Figs. 12 and 13 respectively. Ismail et al. (2022) reported an acceptable accuracy of 92% using their RF model. Using the coupled model, the mean of the average percentage of accuracy for 14 rounds of forecast is $\sim 88\%$ for Gombak (Table 8) and $\sim 84\%$ for Hulu Langat (Table 9). It should be noted that the input data being used for this comparison are not exactly the same. Notably, their study considered epidemiological and environmental factors including notified dengue cases, onset cases, number of interventions, rainfall, temperature, humidity and Air Pollution Index whereas we only focus on temperature, humidity and rainfall. But the proposed coupled model appears to be promising, considering that the percentage of accuracy is above 80%. Even though the coupled model presented a $\sim 4\text{--}8\%$ reduction in accuracy compared to the model proposed by Ismail et al. (2022), the model does not

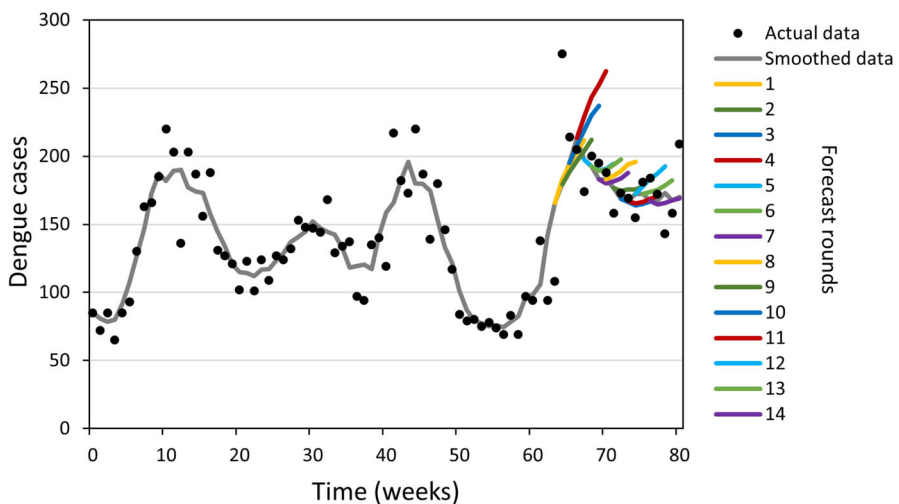


Fig. 12 The forecasted number of dengue cases in Gombak for up to four weeks for fourteen rounds of forecast (forecast starts from Week 65) using the coupled ARIMAX and SI-SIR model

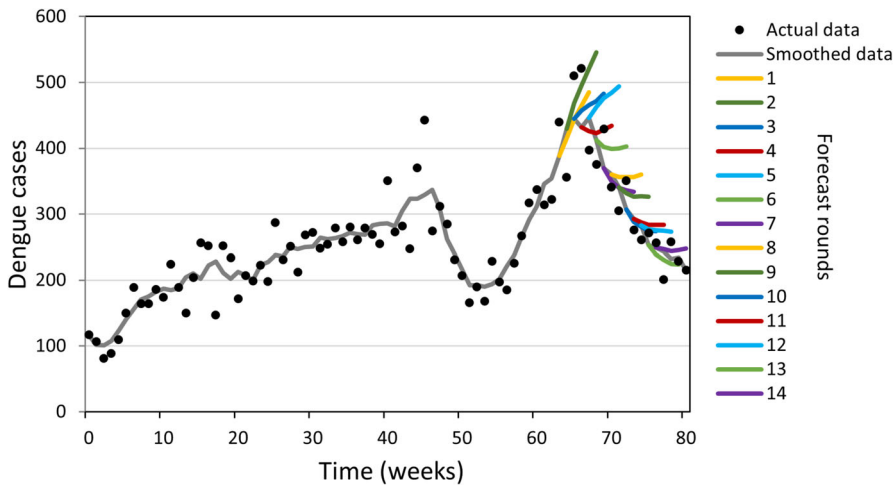


Fig. 13 The forecasted number of dengue cases in Hulu Langat for up to four weeks for fourteen rounds of forecast (forecast starts from Week 65) using the coupled ARIMAX and SI-SIR model

Table 8 Percentage accuracy of forecasted values for Gombak by the coupled model

	Forecasting round						
	1	2	3	4	5	6	7
Average percentage accuracy (%)	83.56	90.23	84.09	68.22	92.29	89.74	90.89
	8	9	10	11	12	13	14
	83.01	95.06	93.58	94.10	88.60	88.05	88.73

Table 9 Percentage accuracy of forecasted values for Hulu Langat by the coupled model

	Forecasting round						
	1	2	3	4	5	6	7
Average percentage accuracy (%)	84.32	77.56	83.07	87.58	65.44	82.76	90.24
	8	9	10	11	12	13	14
	78.60	82.65	94.91	83.28	87.04	90.78	87.26

require entomological data, which can be costly to obtain. Hence, it might be more practical to use the coupled deterministic and ARIMAX model than the RF model reported by Ismail et al. (2022), unless the entomological data is already available. Further, our coupled model is deemed more robust compared to the model proposed by Ismail et al. (2022) because it allows the inclusion of specific intervention measures and additional mosquito-human dynamics into the SI-SIR model to address the specific

needs of a study. Using this coupled model, the effect of various intervention strategies involving single or combined use of chemical approach, biological approach, and new technology on dengue epidemic amid changing climate can be projected.

4 Discussion

Over the past 40 years, dengue incidence in Malaysia has been increasing in frequency and intensity. With no specific treatment and licensed vaccine available, dengue mitigation programs in Malaysia mainly focus on vector eradication and awareness campaign. Hence, the ability to forecast outbreaks would allow public health departments to plan and prepare for impending outbreak. *Aedes* mosquito plays a significant role in dengue transmission (Agha et al. 2017; Wilke et al. 2017). Hence, the forecasting model must consider mosquito population dynamics for better forecasting ability. Climate variables, especially humidity (Lega et al. 2017; Tuladhar et al. 2019), temperature (Haider et al. 2017; Asigau and Parker 2018) and precipitation (Betanzos-Reyes et al. 2018; Nguyen et al. 2020) have been proven to have effects on the mosquito abundance (Asgarian et al. 2021). As the mosquito population increases, their chance of contact with human increases, resulting in higher biting rate (Barnard et al. 1998). Our analysis indicated that there exists a one-way Granger causality between the climate factors and average biting rate, suggesting that climate effects on mosquito population can be collectively simulated through biting rate. Hence, in this study, the three climate variables, namely average temperature, relative humidity and accumulated precipitation, were incorporated as exogenous variables in the ARIMAX model to predict the mosquito biting rate, which in turn was used by the SI-SIR model to forecast the number of infected cases in Selangor for the coming weeks. The simulation result suggests that the coupled model can forecast dengue incidence four weeks ahead with an averaged MMRE of 8.01%. This is important because detecting the onset of the outbreak in advance would allow control measures, such as intensive weekly fogging in hotspot areas, to be planned and implemented to prevent or mitigate the impending outbreak.

When forecasting tools are used make prediction of diseases, there often exist some limitations. Though various forecasting methods have been put forward, many practical applications have shown that epidemic outbreak forecasts often fall far short of the mark, especially for long-term prediction. Like many other time series models, the ARIMAX model performs well in short-term forecasting. Nevertheless, the coupled model appeared to give improved results than the conventional stand-alone statistical models. Apart from the ARIMAX model, multivariate linear and non-linear regression methods have also been tested but the results obtained using these methods are similar to the results produced by the ARIMAX model.

There are several notable limitations to this study. Firstly, the climate effect is incorporated into the model through the mosquito biting rate. However, other rate constants such as mosquito birth rate or mortality rate could also be affected by climate. One of the primary challenges we encountered in this study was the lack of reliable and suitable mosquito data. Hence, introducing another component/function into the model

formulation without data for calibration will increase the uncertainty. Moving forward, it is crucial to obtain mosquito population data to improve model formulation and calibration for a better forecasting capability. Secondly, the SI-SIR model that considers only the basic compartments and one dengue serotype. Incorporating multiple serotypes and more compartments of both mosquito and human population could better capture the dynamics of dengue transmission. However, the required data must be made available to properly calibrate and validate the incorporation of these components. Thirdly, the application case that is limited to a single Malaysian state with prediction accuracy that varies from week to week. The model is less accurate in predicting abrupt turn of events, for instance a sudden jump or drop in dengue cases within a week. But as the updated number is incorporated into the prediction, the model recalibrates to provide a better forecast in the next round. As in the case of many prediction models, the coupled model falls short in terms of predictive accuracy in the early phase of a dengue outbreak. This limitation becomes evident in Table 4, where the first few rounds, corresponding to the early stages of the epidemic, exhibit a relatively high mean magnitude of relative errors compared to the subsequent rounds, which occur during the outbreak. But even with this limitation, our proposed coupled model still outperformed the stand-alone ARIMAX model, particularly at the onset of an outbreak. The predictive capability of this coupled model is not perfect but the forecasting approach proposed here appears to be promising and further application of the model to another two areas in Malaysia, as illustrated in Sect. 3.3, indicated that the model application may not be so limited in its application.

Lastly, the coupled model forecasts better when using biting rates derived from smoothed dengue data compared to its performance when using the real or unsmoothed data. The decision to fit the model to smoothed dengue data was driven by suboptimal forecasting performance observed when using raw data. While this approach offers advantages in capturing the overall trend of dengue cases, it is crucial to recognize that smoothed data deviates from the unaltered observations of real number of cases.

For this proposed method, timely data access for the weekly recalibration is a critical requirement to ensure a good predictive performance. The commendable predictive accuracy in our proposed model is greatly associated with real-time recalibration. Consequently, the model's efficacy relies on the prompt and consistent update of relevant meteorological and dengue data. If the recalibration interval lengthens, the accuracy of the prediction is expected to be reduced. Although the timeliness of official data reporting in Malaysia may vary due to logistical or administrative reasons, weekly or daily retrieval of meteorological and dengue data is feasible, at least, at the ministerial level. Daily retrieval of meteorological data is made possible by MetMalaysia, who has a system to collect and update daily meteorological data, facilitating real-time access to weather forecasts and information. While the MOH Malaysia typically collects and disseminates dengue data regularly, there may be occasional delays in reporting. But as demonstrated during COVID-19 outbreak, the Ministry of Health Malaysia has the ability and platform to capture daily statistics tracking the number of cases, which can be extended to dengue cases. Such feasibility of daily or weekly retrieval of meteorological and dengue data suggests the practicality of the model to be re-calibrated every week.

5 Conclusion

The main objective of the study is to research a method to forecast dengue cases using climate data. Although deterministic models and statistical models have been used individually to forecast dengue cases, forecast by deterministic models often use constant parameters or a fixed function while statistical models are often empirically based. Hence, in this study, we attempt to address such limitations by coupling the statistical ARIMAX and the deterministic SI-SIR models. Such coupling would also allow further research on the effect of intervention strategies on future dengue epidemics subject to the changing climate. In this coupled model, climate was linked to dengue disease through mosquito biting rate, allowing us to study climate effect on dengue cases transmission. The average biting rate time series was first estimated from the curve fitting using SI-SIR model. The Granger causality test performed suggested that the climate variables (average temperature, relative humidity and accumulated precipitation) are useful for predicting the mosquito biting rate. The coupled model was trained using dengue and meteorological data from Jan 2014–May 2019 for Selangor and validated using data from May 2019–Dec 2019. ARIMAX model was used to forecast the average biting rate, which was then used by the SI-SIR model to forecast dengue cases. To capture the drastic change in the number of dengue cases, the model has to be re-calibrated with updated dengue and meteorological data every week. The performance of the coupled model was assessed by calculating the MMRE between the forecasted and recorded number of dengue cases. The model simulation analysis indicated that the average MMRE of 10 rounds stays around 8% for the forecast period (Week 283 to 295). In another randomly chosen validation period (Weeks 204 to 216), the MMRE for all forecast rounds stayed below 12%, indicating that the model forecasting performance is reasonably good irrespective of the validation period chosen. Moreover, the model prediction capability appeared to be fairly consistent, with values of averaged MMRE revolving around 9.63–11%, for the three outbreak periods assessed. When compared to the forecast result using only the ARIMAX model, the forecasted results of the coupled model are deemed more satisfactory, especially at the onset of the outbreak.

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Data availability The dengue and meteorological data that support the findings of this study are available from Malaysian Ministry of Health (MOH) and MetMalaysia, respectively, but restrictions apply to the availability of these data, which were used under licence for the current study, and so are not publicly available. Dengue data are however available from MOH upon application to National Medical Research Register. Meteorological data are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors have no conflicts of interest to declare that are relevant to the content of this article.

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